

## Module 1 Lesson 2

### The Integers; Order of Operations (5.2)

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## Objectives

In this lesson, we will learn how to

- define the integers;
- graph integers on a number line;
- use the symbols  $<$  and  $>$ ;
- find the absolute value of an integer;
- perform operations with integers;
- use the order of operations agreement.

## 1. The Integers

The set consisting of the natural numbers, 0, and the negatives of the natural numbers is called the set of **integers**.

$$\{\dots, -4, -3, -2, -1, 0, 1, 2, 3, 4, \dots\}$$

### Integers

The integers include:

- positive integers;
- zero;
- negative integers.

The term **positive integers** is another name for the natural numbers.

Positive integers can be written in two ways:

+4

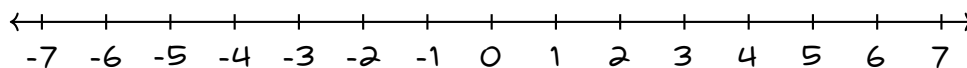
or simply

4.

Both mean "positive four."

## 2. The Number Line

The **number line** is a graph used to visualize the set of integers and other sets of numbers.



### Important Fact

Zero is neither positive nor negative.

Example 1: Graph the following integers on the number line:

-3, 4, 0.

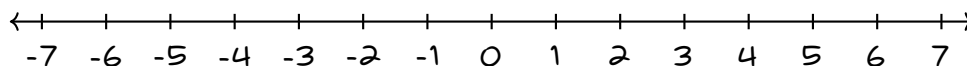
### Solution

The integer -3 is three units to the left of 0.

The integer 4 is four units to the right of 0.

The integer 0 is at the origin.

-3, 0, 4



### 3. The Symbols $<$ and $>$

The symbols  $<$  and  $>$  are called inequality symbols.

Looking at the number line:

$-4$

is to the left of

$-1$ .

This means

$$-4 < -1.$$

Also,

$-1$

is to the right of

$-4$ .

This means

$$-1 > -4.$$

#### Comparing Numbers on a Number Line

Numbers farther to the left are smaller.

Numbers farther to the right are larger.

#### Inequality Symbols

The symbols  $<$  and  $>$  always point to the lesser number when the inequality statement is true.

$$3 < 8$$

$$8 > 3$$

Example 2: Insert either  $<$  or  $>$  between the integers to make each statement true.

$$-4 \square 3$$

$$-1 \square -5$$

$$-5 \square -2$$

$$0 \square -3$$

#### Solution

Compare the locations of the numbers on the number line.

$$-4 < 3$$

$$-1 > -5$$

$$-5 < -2$$

$$0 > -3$$

## 4. Symbols Involving Equality

The symbols  $<$  and  $>$  may be combined with an equal sign.

### Common Inequality Symbols

| Symbol | Meaning                     |
|--------|-----------------------------|
| $<$    | is less than                |
| $>$    | is greater than             |
| $\leq$ | is less than or equal to    |
| $\geq$ | is greater than or equal to |

## 5. Absolute Value

The **absolute value** of an integer  $a$ , denoted by

$$|a|,$$

is the distance from 0 to  $a$  on the number line.

Because absolute value describes a distance, it is never negative.

### Absolute Value

$$|a| = \text{distance from 0 to } a.$$

Therefore,

$$|a| \geq 0.$$

Example 3: Find each absolute value.

$$|-3|$$

$$|5|$$

$$|0|$$

### Solution

Absolute value means distance from 0.

$$|-3| = 3$$

$$|5| = 5$$

$$|0| = 0$$

## 6. Addition of Integers

### Adding Integers with the Same Sign

If the integers have the same sign:

1. Add their absolute values.
2. Keep the common sign.

### Adding Integers with Different Signs

If the integers have different signs:

1. Subtract the smaller absolute value from the larger absolute value.
2. Use the sign of the number with the larger absolute value.

### Study Tip

A good analogy for adding integers is temperature above and below zero on a thermometer.

Think of a thermometer as a number line standing straight up.

Example 4: Add.

$$-8 + 3$$

$$-6 + (-9)$$

$$12 + (-5)$$

Solution

$$-8 + 3 = -5$$

The signs are different, so subtract:

$$8 - 3 = 5.$$

Since 8 has the larger absolute value and  $-8$  is negative, the answer is negative.

$$-6 + (-9) = -15$$

The signs are the same, so add the absolute values and keep the negative sign.

$$12 + (-5) = 7$$

The signs are different, so subtract:

$$12 - 5 = 7.$$

Since 12 has the larger absolute value and is positive, the answer is positive.

## 7. Additive Inverses

Additive inverses have the same absolute value but lie on opposite sides of

zero on the number line.

When we add additive inverses, the sum is 0.

For example,

$$18 + (-18) = 0$$

and

$$(-7) + 7 = 0.$$

#### Additive Inverse Property

For any integer  $a$ ,

$$a + (-a) = 0.$$

## 8. Subtraction of Integers

For all integers  $a$  and  $b$ ,

$$a - b = a + (-b).$$

In words, to subtract  $b$  from  $a$ , add the additive inverse of  $b$ .

The result of subtraction is called the **difference**.

#### Subtracting Integers

To subtract an integer, add its opposite.

$$a - b = a + (-b)$$

Example 5: Subtract.

$$17 - (-11)$$

$$-18 - (-5)$$

$$-18 - 5$$

Solution

To subtract an integer, add its opposite.

$$17 - (-11) = 17 + 11 = 28$$

$$-18 - (-5) = -18 + 5 = -13$$

$$-18 - 5 = -18 + (-5) = -23$$

## 9. Multiplication of Integers

The result of multiplying two or more numbers is called the **product** of the numbers.

Multiplication may be thought of as repeated addition.

Multiplying Two Integers

- The product of two integers with different signs is negative.
- The product of two integers with the same sign is positive.
- The product of 0 and any integer is 0.

Examples:

$$7(-5) = -35$$

$$(-6)(-11) = 66$$

$$-17(0) = 0.$$

### Multiplying Several Integers

If no factor is 0:

- a product with an odd number of negative factors is negative;
- a product with an even number of negative factors is positive.

Example 6: Multiply.

$$(-2)(3)(-4)$$

$$(-1)(-5)(-6)$$

$$(-3)(0)(8)$$

### Solution

$$(-2)(3)(-4)$$

There are two negative factors, so the product is positive:

$$(-2)(3)(-4) = -6(-4) = 24.$$

$$(-1)(-5)(-6)$$

There are three negative factors, so the product is negative:

$$(-1)(-5)(-6) = 5(-6) = -30.$$

$$(-3)(0)(8) = 0$$

Any product with a factor of 0 is 0.

## 10. Exponential Notation

Exponents indicate repeated multiplication.

For example,

$$3^4 = 3 \cdot 3 \cdot 3 \cdot 3.$$

Rules for multiplying integers can be used to evaluate exponential expressions.

### Parentheses Matter

$$(-6)^2 = (-6)(-6)$$

but

$$-6^2 = -(6^2).$$

Example 7: Evaluate.

$$(-6)^2$$

$$-6^2$$

$$(-5)^3$$

$$(-2)^4$$

Solution

$$(-6)^2 = (-6)(-6) = 36$$

$$-6^2 = -(6^2) = -36$$

$$(-5)^3 = (-5)(-5)(-5) = 25(-5) = -125$$

$$(-2)^4 = (-2)(-2)(-2)(-2) = 16$$

## 11. Division of Integers

The result of dividing an integer  $a$  by a nonzero integer  $b$  is called the **quotient**.

We write this quotient as

$$\frac{a}{b}$$

or

$$a/b.$$

For example,

$$\frac{-12}{4} = -3$$

means

$$4(-3) = -12.$$

### Dividing Integers

- The quotient of two integers with different signs is negative.
- The quotient of two integers with the same sign is positive.
- Zero divided by any nonzero integer is zero.
- Division by 0 is undefined.

Example 8: Divide.

$$\frac{-24}{6}$$

$$\frac{-36}{-9}$$

$$\frac{0}{-7}$$

$$\frac{12}{0}$$

### Solution

$$\frac{-24}{6} = -4$$

The signs are different, so the quotient is negative.

$$\frac{-36}{-9} = 4$$

The signs are the same, so the quotient is positive.

$$\frac{0}{-7} = 0$$

Zero divided by any nonzero integer is 0.

$$\frac{12}{0}$$

Division by 0 is undefined, so

$$\frac{12}{0} \text{ is undefined.}$$

## 12. Order of Operations

The **order of operations** tells us which operations to perform first when simplifying an expression.

### Order of Operations

1. Perform all operations within grouping symbols.
2. Evaluate all exponential expressions.
3. Do all multiplications and divisions in the order in which they occur, working from left to right.
4. Do all additions and subtractions in the order in which they occur, working from left to right.

Example 9: Simplify.

$$6^2 - 24 \div 2^2 \cdot 3 + 1.$$

#### Solution

First evaluate exponents:

$$6^2 = 36$$

and

$$2^2 = 4.$$

So,

$$6^2 - 24 \div 2^2 \cdot 3 + 1 = 36 - 24 \div 4 \cdot 3 + 1.$$

Next perform division and multiplication from left to right:

$$24 \div 4 = 6$$

$$6 \cdot 3 = 18.$$

So,

$$36 - 24 \div 4 \cdot 3 + 1 = 36 - 18 + 1.$$

Finally, perform addition and subtraction from left to right:

$$36 - 18 + 1 = 18 + 1 = 19.$$

Thus,

$$\boxed{19}$$

## Summary

### Key Ideas

- The integers are

$$\{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}.$$

- Zero is neither positive nor negative.
- Numbers to the left on the number line are smaller.
- Numbers to the right on the number line are larger.
- Absolute value represents distance from 0.
- To subtract an integer, add its opposite.
- A product or quotient of integers with the same sign is positive.
- A product or quotient of integers with different signs is negative.
- Division by 0 is undefined.
- Use the order of operations to simplify expressions correctly.